QUIZ 2 -MTH301-page5-8

Graded

Student

YATIN PREETAM BHOJWANI

Total Points

7 / 10 pts

Question 1

Q3 — 0 / 3 pts

- ✔ **+ 0 pts** Completely incorrect/ Not attempted
- **+ 3 pts** Completely Correct

Question 2

Q4 — 3 / 3 pts

- **+ 0 pts** Completely incorrect/ Not attempted
- ✔ **+ 3 pts** Completely Correct

Question 3

Q5 — 4 / 4 pts

- **+ 0 pts** Completely incorrect/ Not attempted
- ✔ **+ 4 pts** Completely Correct
- **+ 2 pts** Only one part is correct

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3. (M, d) is a connected metric space and has at-least two elements. Show that M is uncountable. [3]

(M,d) has at least two elements, ~~i.e $a, b \in M$ & $a \neq b$.~~ a_1, a_2

i.e $a_1, a_2 \in A$ & $a_1 \neq a_2$

Proof by contra positive:

To show if M is atmost countable then M is disconnected.

let M be atmost countable.

i.e. $M = \cancel{\{a, b, c, \ldots\}}$

$M = \{a_i : i \in N\}$

let $A = \{a_1\}$

$B = \{a_2, a_3, \ldots\}$

$A \cap B = \phi$

and $A \cup B = M$

$\Rightarrow M$ is disconnected.

(This includes case of finite but ~~let me at~~ more elaboration below)

Let M be finite.

$M = \{a_i : 1 \leq i \leq n\}$ $n \geq 2$

let $A = \{a_1\}$

$B = \{a_2, \ldots a_n\}$

again $A \cap B \neq \phi$

4. Let (M, d) be a metric space and $A \subseteq M$. Prove that if A is totally bounded then (the closure) $\overline{A}$ is totally bounded. [3]

A is totally bounded.

$\Rightarrow \forall \varepsilon > 0$,

$\exists$ $x_1, x_2, \ldots x_n$

s.t $A \subseteq \bigcup_{i=1}^{n} B_{\varepsilon}(x_i)$

to show $\forall \varepsilon > 0$

$\exists y_1, y_2, \ldots y_m$

s.t $\overline{A} \subseteq \bigcup_{i=1}^{m} B_{\varepsilon}(y_i)$

~~take $y \in$~~

$\forall \varepsilon > 0$

for $x \in \overline{A}$ $(A \subseteq \bigcup_{i=1}^{n} B_{\varepsilon/2}(x_i))$

$B_{\varepsilon/2}(x) \cap A \neq \phi$

$B_{\varepsilon/2}(x) \cap \bigcup_{i=1}^{n} B_{\varepsilon/2}(x_i) \neq \phi$

$\Rightarrow B_{\varepsilon/2}(x) \cap B_{\varepsilon/2}(x_i) \neq \phi$

for some $1 \leq i \leq n$

~~take~~. $\exists y$ s.t

$y \in B_{\varepsilon/2}(x)$ &

$y \in B_{\varepsilon/2}(x_i)$

$d(x_i, x_i) \leq d(x, y) + d(x_i, y)$

$< \varepsilon$

$\Rightarrow d(x, x_i) < \varepsilon$

$x \in B_\varepsilon(x_i)$ ...

$\Rightarrow \overline{A} \subseteq \bigcup_{i=1}^{n} B_{\varepsilon}(x_i)$

$\Rightarrow \overline{A}$ is totally bounded.

$A \cup B = M$

$\Rightarrow M$ is disconnected.

$\Rightarrow x \in N_\varepsilon(x_i)$ for some $1 \leq i \leq n$

5. Let (M, d) be a complete metric space. A subset $A \subseteq M$ is complete if and only if A is a closed set.

[4]

M is a complete metric space.

$\Leftrightarrow$ any cauchy sequence (x_n) in M converges to $x \in M$.

A subset $A \subseteq M$ is complete.

if any cauchy sequence (x_n)

A is a closed set.

and (M, d) is a complete metric space.

~~consid~~

~~any~~ consider any

cauchy sequence (x_n)